

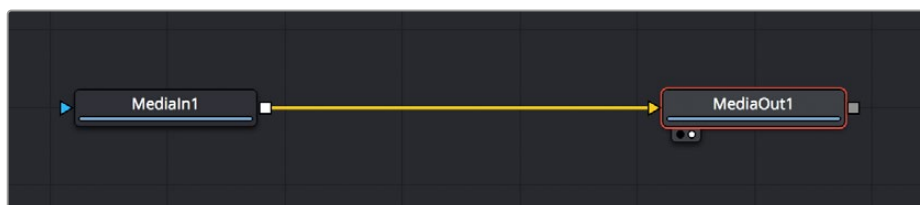
Preparing Compositions in the Fusion Page

Ordinarily, clips come into the Fusion page from the Edit or Cut page Timeline as either a single clip, or as multiple layers contained within a Fusion clip. You can also add clips to a composition directly from DaVinci Resolve's Media Pool. How clips find their way into a Fusion composition can determine how they function within that composition and what resolution that composition outputs to the rest of DaVinci Resolve.

Working on Single Clips in the Fusion Page

Positioning the playhead over a clip in the Edit page or Cut page Timeline and clicking the Fusion page button causes that clip to appear in the Fusion page as a single MediaIn node connected to a MediaOut node. Only the topmost visible clip is taken into Fusion. Clips that aren't visible because they're on lower tracks with clips above them are ignored, unless you disable the clips or tracks that appear above. These very simple default compositions are referred to unofficially in this manual as "single-clip compositions."

The MediaIn node represents the image that's fed to the Fusion page for further work, and the MediaOut node represents the final output that's fed onward to the Color page for grading.



The default node tree that appears when you first open the Fusion page while the playhead is parked on a clip.

This initial node structure makes it easy to quickly use the Fusion page to create relatively simple effects using the procedural flexibility of node-based compositing.

For example, if you have a clip that's an establishing shot, with no camera motion, that needs some fast paint to cover up a bit of garbage in the background, you can open the Fusion page, add a Paint node, and use the Clone mode of the Stroke tool to paint it out quickly.



A simple paint effect applied to a shot with no camera motion.

TIP: The resolution of a single clip brought into Fusion via the Edit or Cut page Timeline is the resolution of the source clip, not the Timeline resolution.

Once you've finished, simply go back to the Edit or Cut page and continue editing, because the entire Fusion composition is encapsulated within that clip, similarly to how grades in the Color page are also encapsulated within a clip. However you slip, slide, ripple, roll, or resize that clip, the Fusion effects you've created and the Color page grades you've made follow that clip's journey through your edited Timeline.

TIP: While you'll likely want to do all the compositing for a greenscreen style effect in the Fusion page, it's also possible to add a keyer, such as the excellent DeltaKeyer node, between the MediaIn and MediaOut nodes, all by itself. When you pull a key this way, the alpha channel is added to the MediaOut node, so your clip on the Edit page has transparency, letting you add a background clip on a lower track of your Edit page Timeline.

How Nodes Are Named

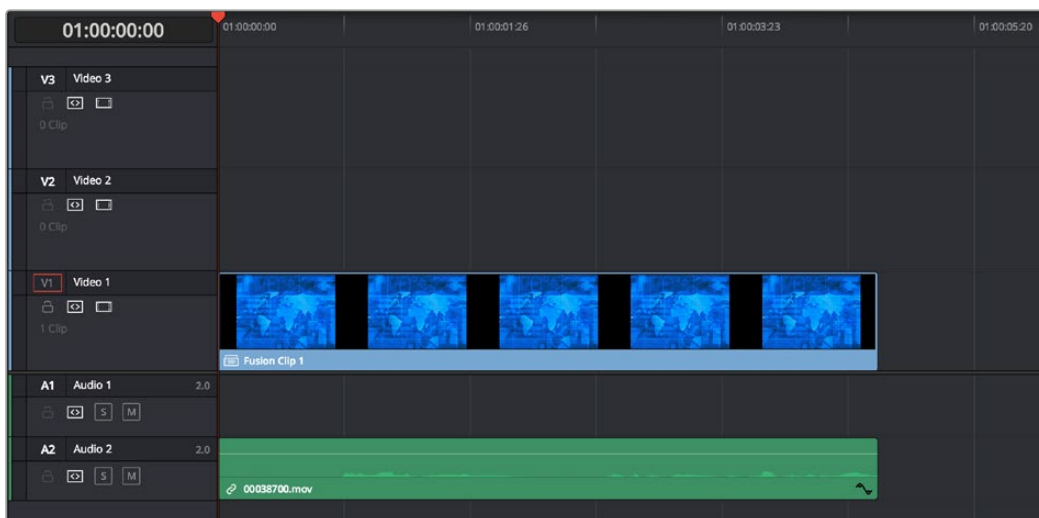
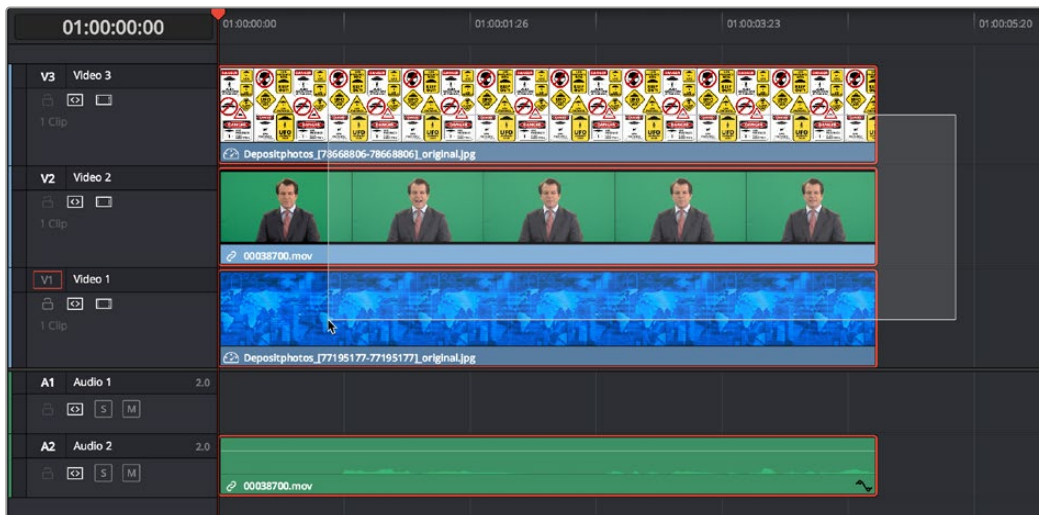
While the documentation refers to nodes by their regular name, such as "MediaIn," the actual names of nodes in the Fusion Node Editor have a number appended to them, to indicate which node is which when you have multiple instances of a particular type of node.

Turning One or More Clips into Fusion Clips

For situations where you know you're creating a more ambitious composited effect that requires multiple layers edited together with very specific timings, you can create a "Fusion clip" right from the Timeline. For example, if you have a foreground greenscreen clip, a background clip, and an additional graphic clip, you can stack them all on the Timeline as superimposed clips. You can then use the Edit page slip and slide features to align their timings so they work together as necessary. You can also edit multiple consecutive clips together that you want to use in a composition as a series of clips. Once that's done, you can select every clip in the stack to create a Fusion clip, so you can easily use all these superimposed layers within a Fusion composite.

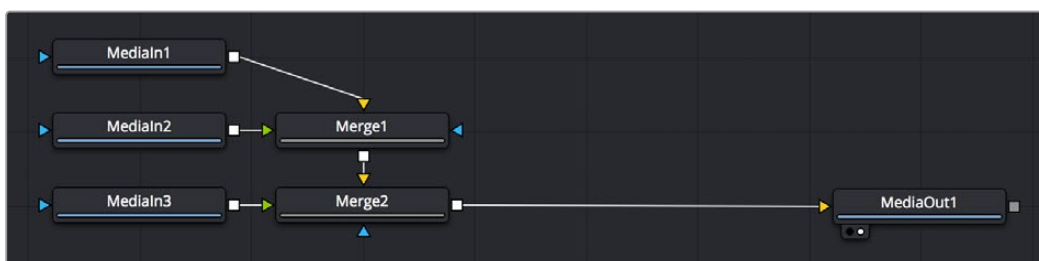
To create a Fusion clip:

- 1 Edit all the clips you want to use in the Edit page Timeline.
- 2 Select all clips you want to be in the same composite at once.
- 3 Right-click one of the selected clips and choose New Fusion Clip from the contextual menu.
- 4 A new clip, named "Fusion Clip X" (where X is an automatically incrementing number) appears in the currently selected bin of the Media Pool and in the Timeline to replace the previously selected clips.
- 5 With the playhead parked over that clip, open the Fusion page to see the new arrangement of those clips in the Fusion page Node Editor.



A stack of clips to use in a composite (Top), and turning that stack into a Fusion clip in the Edit page (Bottom).

The nice thing about creating a Fusion clip is that every superimposed clip in a stack is automatically connected into a cascading series of Merge nodes that create the desired arrangement of clips. Note that whatever clips were in the bottom of the stack in the Edit page appear at the top of the Node Editor in the Fusion page, but the arrangement of background and foreground input connections is appropriate to recreate the same compositional order.



The initial node tree of the three clips we turned into a Fusion clip.

TIP: Fusion clips change the working resolution of the individual clips to match the Timeline resolution. For instance, if two 4K clips are stacked one on top of the other in an HD Timeline, creating a Fusion clip resizes the clips to HD. The full resolution of the individual 4K clips is not available in Fusion. To maintain the full resolution of source clips, bring only one clip into the Fusion composition from the Edit or Cut page Timeline, and then bring other clips into the Fusion composition using the Media Pool.

Fusion Referenced Compositions

There is a newer powerful way of applying Fusion composites to your clips called a “Referenced Composition.” This method allows you to reuse your work across multiple clips or even timelines.

You create a referenced composition on the Edit page by selecting a single clip, or a stack of clips, and then select “Create Referenced Composition” from the contextual menu. This generates a Fusion Referenced Composition in the Media Pool. You can enter the Fusion page to start compositing or double-click the Referenced Composition in the Media Pool to open it.

Referenced comps can be linked to other clips in any timeline by selecting the referenced comp, then right-clicking the target clip and selecting “Link to Referenced Composition.” Any changes you make are applied across all connected clips because they are all “referred” or “linked” to the same composition.

Referenced Compositions are an effective and secure way to keep your work backed up and organized, as they live in the Media Pool and will persist even if the original referenced clip is deleted from the timeline.

Creating and Managing Referenced Compositions

To create a Referenced Composition:

- 1 Select one or more timeline clips that are “stacked” on tracks in an Edit window timeline (clips have to overlap in time at least partially).
- 2 Right-click and select “Create Referenced Fusion Composition.”
- 3 The new reference composition, associated with the top selected clip, now appears in the Media Pool.

You can open the referenced composition in the Fusion page by double-clicking on it in the Media Pool. Media nodes match the relative locations of the source track clip arrangement in the Edit timeline.

The contents of the video track are treated as a source, allowing creative manipulation of clips before they’re passed on to Fusion for compositing.

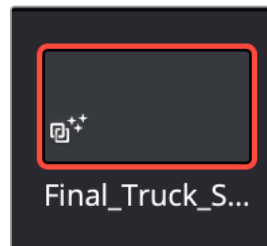
You can right-click on a referenced composition in the Media Pool and choose the Usage option to view and navigate to its target linked clips.

You can duplicate your referenced composition for iteration purposes.

In the example below we create a simple reference composition out of three clips, replacing a logo on a truck.

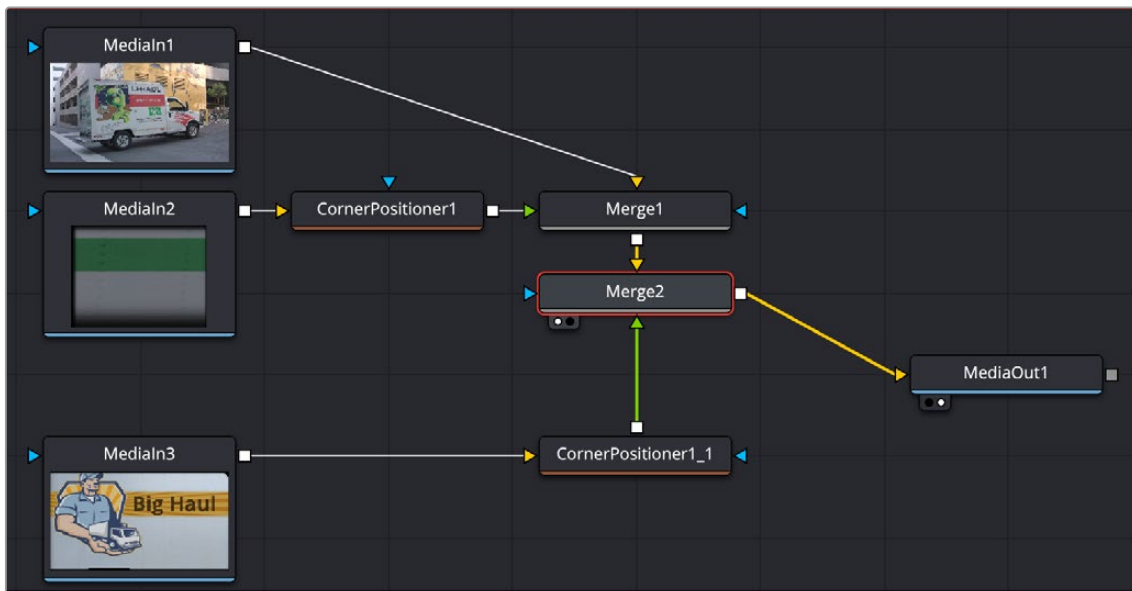


Creating a reference composition by selecting multiple overlapping clips, and then Create Referenced Composition from the contextual menu.

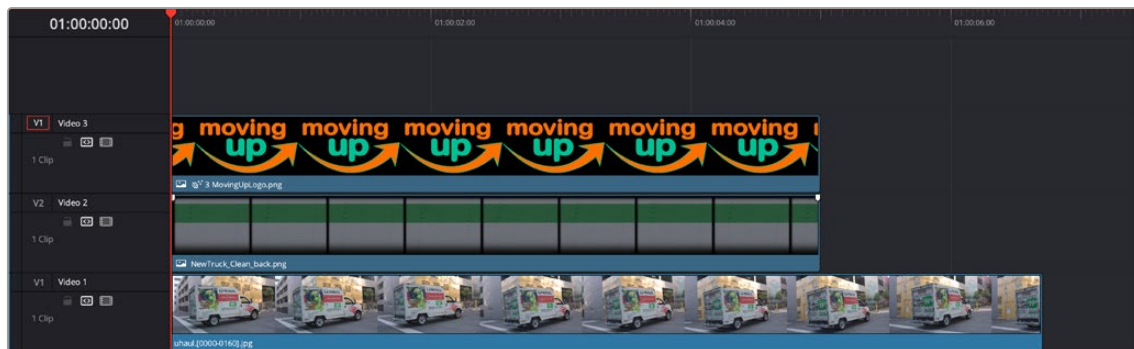


The Reference Composition thumbnail in the Media Pool

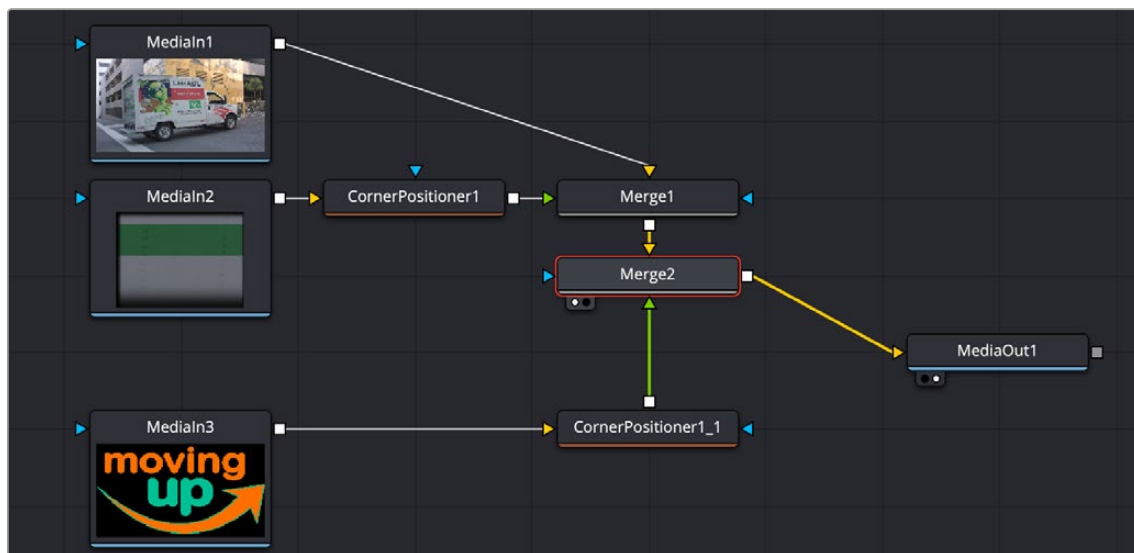
Double-clicking on the Reference Node brings us into Fusion, where we add some Corner Positioner nodes to line up the green stripe and logos to fit.



Now that we have the referenced composition set, we can easily change the logo, just by performing a replace edit with the new logo to V3 in the timeline. The new logo is automatically applied in the Fusion comp with the same settings.



The sBSpline controls



Adding Additional Referenced Clips

Once you've established a referenced composition, you can add clip(s) that can use the referenced composition to affect the other clips.

To add clip references to a referenced composition:

- 1 Select the referenced composition that you want to refer to another clip.
- 2 Select a target clip in the timeline, right click and choose "Link to Referenced Composition."
- 3 Add other clips, one a time if you wish, including clips from other timelines.

Referenced Composition Behaviors and Rules

There are some additional behaviors and rules to be aware of when using referenced compositions.

- Referenced compositions maintain the original source resolution of your clips.
- When selecting multiple clips, ensure that the clips have some overlap with the top selected clip. Without this overlap, the context menu action to create a referenced composition across layers is not available.
- When creating a project composition from multiple overlapping clips, a single composition associated with the top selected clip is created.
- The referenced composition features additional Media In nodes (one per track). Media In 1 refers to the top selected clip. Subsequent Media In nodes reference timeline tracks (layers) below the top clip.
- Removing a timeline clip will not remove the associated composition.
- When you edit the contents in a referenced composition's track, the corresponding Media In refreshes to automatically to show the new clips, edits, effects, and transitions.

From the timeline clip's context menu, you can:

- **Reset Reference Composition:** To reset the composition. This affects all referenced clips.
- **Unlink Reference Composition:** To remove the composition reference from the selected clip. This retains the composition in the Media Pool, but the clip effect is cleared. Useful if you want to link to another referenced composition.
- **Find Reference Composition in Media Pool:** To find the reference composition the clip is a part of in the Media Pool.

NOTE: Clips of differing frame rates or clips with speed changes are not supported in referenced compositions.

Adding Fusion Composition Generators

The Generator category of the Edit page Effects Library has a Fusion Composition generator. It's useful for creating an empty placeholder in the Timeline that you later want to work on in the Fusion page to create a more fully-featured Fusion composition.

To create a blank Fusion clip in the Edit page:

- 1 Open the Effects Library, and select the Effects category.
- 2 Edit a Fusion Composition clip into the Timeline in whichever way is most convenient.
 - a) You can drag a Fusion Composition clip into the Timeline, which will result in a clip that's the length of the "Standard generator duration" preference, which is 5 seconds by default.
 - b) You can set In and Out points in the Timeline, and drag the Fusion Composition clip onto any of the editing overlays of the Timeline viewer to perform that sort of edit to insert, overwrite, "place on top," or ripple overwrite it into a specific place in the Timeline, for a specific duration.
- 3 A new clip named "Fusion Composition" appears in the Timeline. It initially displays only black in the Timeline viewer, since it's a blank composition with no contents.
- 4 With the playhead parked over that clip, open the Fusion page. Since this composition is blank, there's only a single MediaOut node in the Node Editor. At this point, you can add whatever media, generators, and other effects you require.

Creating a Fusion Composition Clip in a Bin

You can create an empty Fusion Composition clip in any bin in the Media Pool without creating a Timeline. This method can be useful for creating motion graphics or titles when there is no Timeline or when you plan on using the clip in multiple Timelines.

To create a blank Fusion Composition clip in a bin:

- 1 Select the bin in the Media Pool where you want to save the Fusion Composition.
- 2 Right click in an empty area of the bin and choose New Fusion Composition.
- 3 In the New Fusion Composition clip dialog, enter a Name for the clip, a duration, and a frame rate, and then click Create.
- 4 The clip will appear in the bin. To open it in Fusion, do one of the following:
 - Double-click the Fusion Composition
 - Right-click over the Fusion Composition clip and choose Open in Fusion Page

Resetting a Fusion Clip

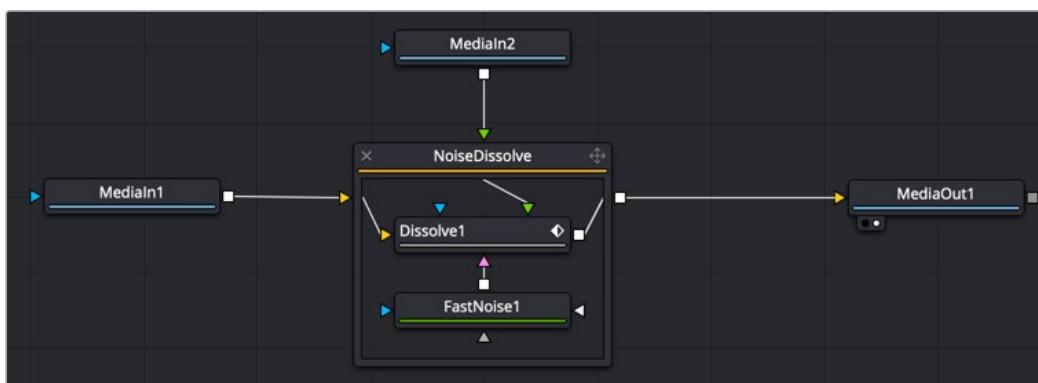
To reset a Fusion clip back to its initial default state, right-click on the Fusion clip in the Edit page Timeline, and select Reset Fusion Composition. You can select multiple Fusion clips and reset them at once.

Using Fusion Transitions

Specific Fusion transitions are available in the Edit page Effects Library. You can use these transitions to bring two clips and the transition between them into the Fusion Page. These transitions can then be modified and saved back to the Edit page Timelines or saved as a new, reusable Fusion Transition that appears in the Edit Page Effects Library.

To apply and open a Fusion Transition:

- 1 Open the Effects Library, and select the Video Transitions category.
- 2 Scroll to the bottom of the transition list and drag one of the Fusion Transitions onto a cut in the Timeline.
- 3 Right-click over the Fusion Transition in the Timeline and choose Open in Fusion Page.
- 4 The Fusion page opens with two MediaIn nodes representing the two sides of the transitions. The MediaIn nodes connect to a cross dissolve or a group of nodes used to create the transition. At this point, you can modify the transition using masks or other nodes and return to the Edit page to see the results.



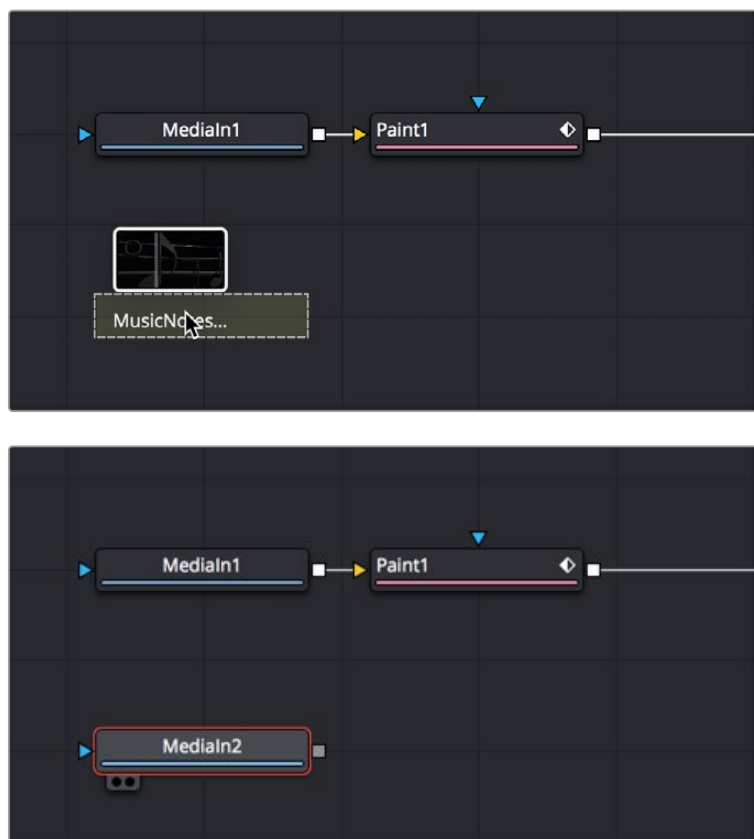
A Noise Dissolve Transition opened in the Fusion page.

To learn about creating custom Fusion Transitions that appear in the Effects Library, see *Chapter 68, “Node Groups, Macros, and Fusion Templates,”* in the DaVinci Resolve Reference Manual or Chapter 6 in the Fusion Reference Manual.

Adding Clips from the Media Pool

You can open the Media Pool on the Fusion page and drag clips directly to the Node Editor to add them to your node tree.

When you add a clip by dragging it into an empty area of the Node Editor, it becomes another MediaIn node, disconnected, that’s ready for you to merge into your current composite in any one of a variety of ways.



Dragging a clip from the Media Pool (Top), and dropping it onto your composition (Bottom).

TIP: Dragging a clip from the Media Pool on top of a connection line between two other nodes in the Node Editor adds that clip as the foreground clip to a Merge node.

When you add additional clips from the Media Pool, those clips become a part of the composition, similar to how Ext Matte nodes you add to the Color page Node Editor become part of that clip’s grade.

Audio with Media Pool Clips

Audio from a clip brought in through the Media Pool is muted by default. Hearing the audio from a Media Pool clip is a two step process.

To hear audio from a clip brought in through the Media Pool, do the following:

- 1 Select the clip in the Node Editor.
- 2 In the Inspector, click the Audio tab and select the clip name from the Audio Track drop-down menu.
- 3 Right-click the speaker icon in the toolbar, then choose the MediaIn for the Media Pool clip to solo its audio.

You can now use the speaker icon contextual menu to switch back and forth between all the MediaIn nodes.

Finding Clips in the Media Pool from the Node Editor

Complex Fusion Compositions can have multiple MediaIn nodes to keep track of. It is possible to find the exact clip in the Media Pool assigned to a MediaIn node by selecting it, and then choosing Clip > Find Clip in Media Pool (Option-F).

Adding Clips from the File System

You also have the option of dragging clips from the file system directly into the Node Editor. When you do this, they'll be added to the currently selected bin of the Media Pool automatically. So, if you have a library of stock animated background textures and you've just found the one you want to use using your file system's search tools, you can simply drag it straight into the Node Editor to use it right away.

Using MediaIn Nodes

The MediaIn nodes in the Fusion Page are the foundation of every composition you create. This section provides more detail about the controls available for adjusting MediaIn and Loader nodes.

MediaIn Node Inputs

MediaIn nodes have one Effects mask input and one output. In the case of the Effects mask input, connecting a mask node such as a Polygon or B-Spline node creates an alpha channel in the MediaIn node.

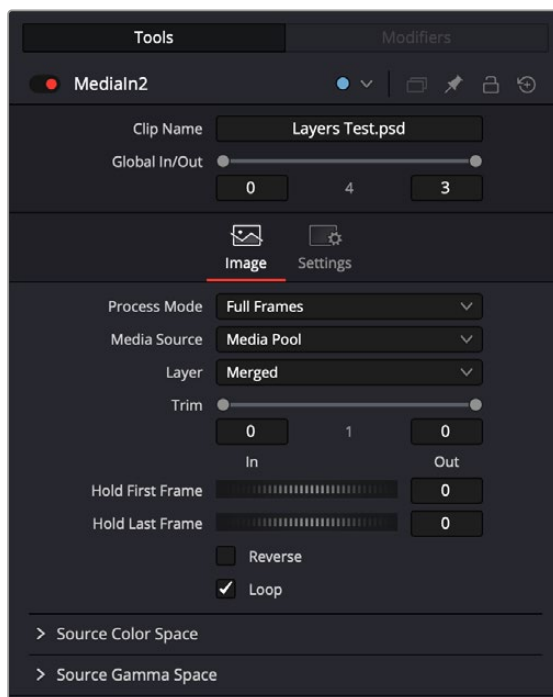
TIP: If you connect a mask node without any shapes drawn, that mask outputs full transparency, with the result that the image output by the MediaIn node is uselessly blank. If you want to rotoscope over a MediaIn node, first create a disconnected mask node, and with the mask node selected (exposing its controls in the Inspector) and the MediaIn node loaded into the viewer, draw your mask. Once the shape you're drawing has been closed, you can connect the mask node to the MediaIn node's input, and you're good to go.

Inspector Properties of MediaIn Nodes

Which Inspector options are available for MediaIn nodes of your composition depends on how you imported the media.

MediaIn Node Parameters for Clips in a Timeline

When you create a composition using clips from the Edit page Timeline, the MediaIn nodes for those clips display fewer parameters than those imported directly from the Media Pool, because the timing of clips that have already been edited into a Timeline is already set.



The Inspector Image tab parameters for a clip from the Timeline

Image Tab

- **Clip Name:** Displays the name of that clip.
- **Process Mode:** Lets you choose whether the clip represented by that node will be processed as Full Frames, or via one of the specified interlaced methods.
- **Media Source:** This control allows you to choose where media is sourced from; the three available options include: Timeline, Background, and Media Pool.
 - Timeline:** Links the clip from the Edit Timeline (default).
 - Background:** This pulls frames from the lower video tracks in the edit. When toggling the Media Source to Background, the composited result of all the lower tracks is visible.
- **Media Pool:** Links the clip directly from the Media Pool, bypassing the Edit Timeline.
- **Layer:** In the case of Multilayer or Photoshop PSD files, this selects the layer in the file to use. For a Fusion Clip created using multiple video layers on the timeline, this will refer to the index number of each of the layers.

Source Color Space

The Color Space Type menu sets the color space of the media to help achieve a linear workflow. Unlike the Gamut tool, this doesn't perform any actual color space conversion but rather adds the source space data into the metadata, if that metadata doesn't already exist. The metadata can then be used downstream by a Gamut tool with the From Image option, or in a Saver if explicit output spaces are defined there.

- **Auto:** Passes along any metadata that might be in the incoming image.
- **Space:** Allows the user to set the color space from a variety of options.

Source Gamma Space

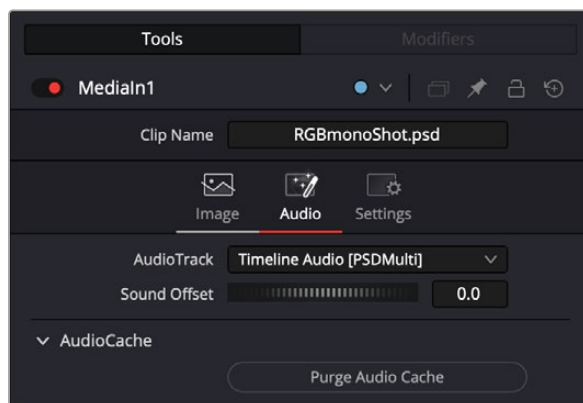
The Curve Type menu automatically determines or allows you to choose the Gamma setting for the image and allows the option to remove the Gamma curve to help achieve a linear workflow.

- **Auto:** Passes along any metadata that might be in the incoming image.
- **Space:** Allows you to choose a specific setting from a Gamma Space drop-down menu, while a visual graph lets you see a representation of the gamma setting you've selected.
- **Log:** Similar to the Log-Lin node, this option reveals specific log-encoded gamma profiles so that you can select the one that matches your content. A visual graph shows a representation of the log setting you've selected. When Cineon is selected from the Log Type menu, additional Lock RGB, Level, Soft Clip, Film Stock Gamma, Conversion Gamma, and Conversion table options are presented to finesse the gamma output.
- **Remove Curve:** Depending on the selected gamma space or on the gamma space found in Auto mode, the associated gamma curve is removed from the material, effectively converting it to output in a linear color space.
- **Pre-Divide/Post-Multiply:** Lets you convert "straight" alpha channels into pre-multiplied alpha channels, when necessary.

TIP: All content in the DaVinci Resolve Fusion page is processed using 32-bit floating-point bit depth, regardless of the content's actual bit depth.

Audio Tab

The Inspector for the MediaIn node contains an Audio tab, where you can choose to solo the audio from the clip or hear all the audio tracks in the Timeline.



The MediaIn Audio tab

The Audio tab in the MediaIn node is used to select the track for playback, slip the audio timing, and reset the audio cache

If the audio is out of sync when playing back in Fusion, the Audio tab's Sound Offset wheel allows you to slip the audio in subframe frame increments. The slipped audio is only modified in the Fusion page. All other pages retain the original audio placement.

Purging the Audio Cache

The audio and its settings are cached for faster performance. If you change which audio tracks you want to play back in Fusion, or you use the Sound Offset wheel to slip the audio tracks, you need to purge the audio cache. Also, if you return to the Edit, Cut, or Fairlight page and modify the audio levels, you need to purge the audio cache.

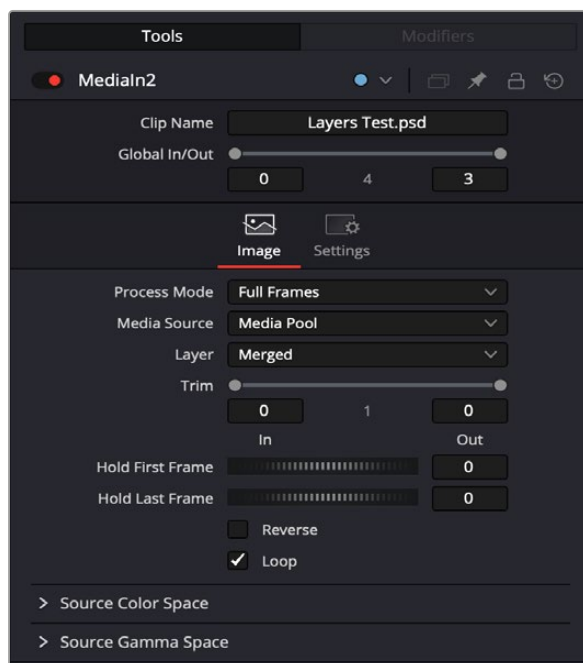
To purge the audio cache after any change to the audio playback:

Click the Purge Audio Cache button in the Inspector.

The audio will be updated when you next play back the composition.

Aligning Clips from the Media Pool

When you add a clip from the Media Pool or your file system directly into a composition, the resulting MediaIn node has more options than clips from the Edit page Timeline. These additional options make it easier to align the Media Pool clips with other clips from the Edit or Cut page Timeline. You can also trim clips, hold the first or last frame for a longer duration than the original media, and reverse or loop the clip to get more range for your composition.



The Inspector parameters for a clip imported from the Media Pool.

Below is a list of controls that are added beyond the controls that appear when a clip is added from the Edit or Cut page Timeline.

- **Global In/Out:** Use this control to specify the position of this node within the composition. For instance, when the clip is added to the comp from the Media Pool, it is added at frame 0. However, the MediaIn node from the Edit page Timeline may not start until a much later frame, based on where it is edited into the Timeline. Use Global In to specify the frame on which that the clip starts so that it aligned with media from the Edit page Timeline. It is easiest to view and change the alignment of different clips in the comp while viewing the Keyframes Editor. To slide the clip in time or align it to other clips without changing its length, place the mouse pointer in the middle of the range control and drag it to the new location, or enter the value manually in the Global In value control. If the Global In and Out values are decreased to the point where the range between the In and Out values is smaller than the n number of available frames in the clip, Fusion automatically trims the clip by adjusting the Clip Time range control. If the Global In/Out values are increased to the point where the range between the In and Out values is larger than the number of available frames in the clip, Fusion automatically lengthens the clip by adjusting the Hold First/Last Frame controls. Extended frames are visually represented in the range control by changing the color of the held frames to purple in the control.
- **Trim:** The Trim range control is used to trim frames from the start or end of a clip. Adjust the Trim In to remove frames from the start and set Trim Out to specify the last frame of the clip. The values used here are offsets. A value of 5 in Trim In would use the 5th frame in the sequence as the start, ignoring the first four frames. A Trim Out value of 95 would stop loading frames after the 95th.
- **Hold First Frame/Hold Last Frame:** The Hold First Frame and Hold Last Frame controls will hold the first or last frame of the clip for the specified amount of frames. Held frames are included in a loop if the footage is looped.
- **Reverse:** Select this checkbox to reverse the footage so that the last frame is played first and the first frame is played last.
- **Loop:** Select this checkbox to loop the footage until the end of the project. Any lengthening of the clip using Hold First/Last Frame or shortening using Trim In/Out is included in the looped clip.

Using Loader and Saver Nodes in the Fusion Page

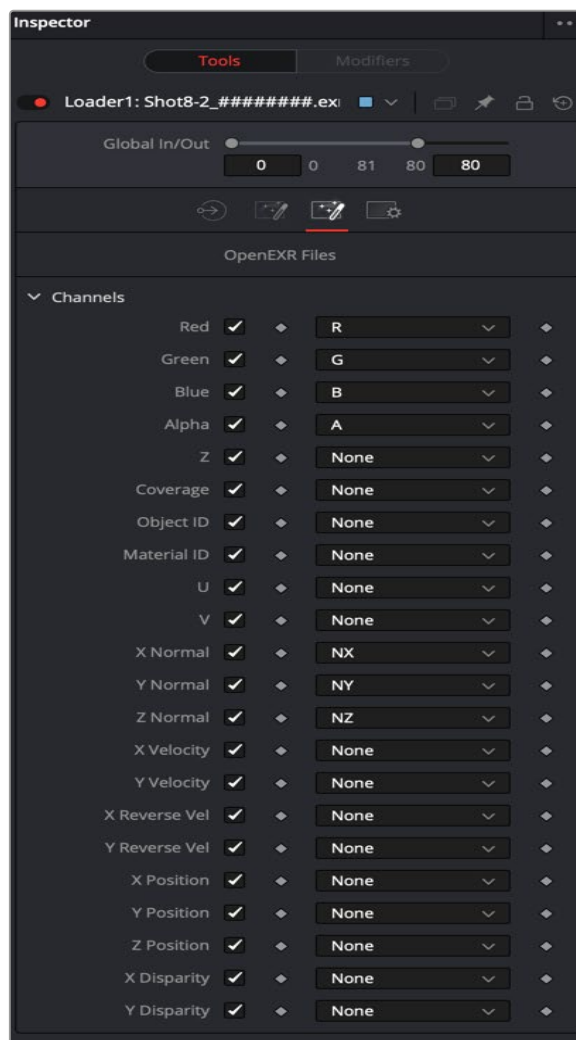
The Loader and Saver nodes in Fusion Page are specifically used for workflows that center around multi-channel EXR files. OpenEXR media can contain high-quality floating-point image data, multiple matte channels, as well as auxiliary channels rendered from 3D software. The Loader node lets you add OpenEXR files to a composition directly from the file system, retaining all the channels that are embedded within that file. Saver nodes enable you to render either all or part of a composition as EXR files directly to disk, bypassing the DaVinci Resolve Deliver page.

To import a multi-channel EXR image sequence into the Fusion page:

- 1 Open the Fusion page.
- 2 Open the Effects Library, and select the Tools > I/O category, and click the Loader node.
- 3 In the OS navigation window that opens, select the EXR image sequence you want to import, and click Open.
- 4 A Loader node linked to the EXR file will appear in the Fusion page, although a clip will not appear in the Media Pool.

Loader Node Parameters

The image tab of Loader nodes shares parameters with MediaIn nodes, as described previously in this chapter. However, when using a Loader node for EXR files, the Format tab selectively enables and disables the use of specific auxiliary channels contained in the file.



The Format tab in a Loader node Inspector displays Aux channels in EXR files.

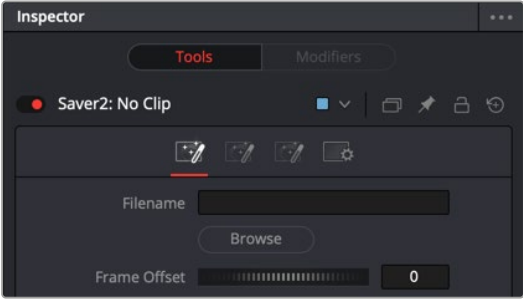
Outputting Images Using Saver Nodes

Saver nodes render OpenEXR image sequences to disk directly from the Fusion page. Saver nodes can be added to any branch of a node tree, allowing you to export one or more subsets of nodes in a composition. You can add as many Saver nodes as you like to whichever branches of your composition's node tree you need to output, to export multiple parts of a composition. For example, if a particular composition is made up of one branch of nodes working together to create a complex background, and another branch of nodes creating a foreground character with transparency, you could export the background and foreground branches as separate OpenEXR files using two Saver nodes.

To do this, simply create a Saver node after any set of nodes you want to output, and then open the Inspector and click Browse to choose a name and location for the rendered result.

When naming the file, you must add the .exr file extension. Fusion sets the output format accordingly. A four-digit frame number is automatically added before the filename extension. However, you can specify the frame padding by adding several zeroes to indicate the number of digits. For example, 000 signifies 001.

Once the Saver node is set up, output one or more Saver nodes, and choose **Fusion > Render All Savers**.



The Inspector parameters for a Saver node.

Manual Disk Caching Using Loader and Saver Nodes

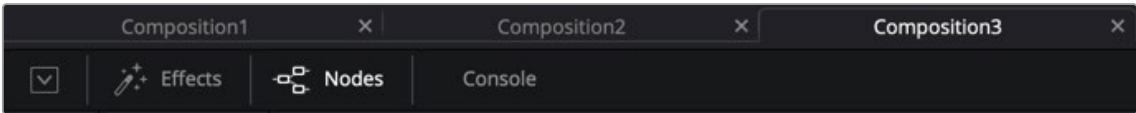
The Loader and Saver nodes in the Fusion page are also useful for optimizing extremely complex and processor-intensive compositions. For example, you can render out specific branches of a node tree that no longer require frequent adjustment to OpenEXR via a Saver node, then reimport the result to take the place of the original branch of nodes in order to improve the performance of your composition. Used this way, Loader and Saver nodes provide a bulletproof manual workflow for caching using media files that will never be automatically purged unless you specifically delete them. As long as you retain the original branch of nodes, you can always readjust and re-render these manually cached parts of a composition, if necessary.

Preparing Compositions in Fusion Studio

The next few sections in this chapter cover preparing a project and adding clips into a composition when using Fusion Studio. The term composition, or comp, is used to refer to the Fusion project file. By default, opening the Fusion Studio application creates a new empty composition when it's launched. A composition can contain single frames, image sequences, or movie files at various resolutions and bit depths. Knowing which files you can load in, how to set up a composition to handle them, and finally, reading those files in are the first steps in beginning to composite.

Opening, Closing, and Saving Compositions

As soon as you open Fusion Studio, a new empty composition is created. If necessary, you can also create or open multiple compositions at once. Each additional composition is opened as a tab to the main Fusion Studio window.



Three compositions opened as tabs in Fusion Studio.

To create a second new composition:

- Choose **File > New**.

To open an existing composition, do one of the following:

- Choose File > Open.
- Choose File > Open Recent, and choose from the list of recently opened comps.
- Drag a composition file from an OS file browser into the tabbed composition area at the top of the Fusion Studio window.
- Double-click on a composition file in the OS file browser.

The following methods can be used to close the current composition:

- Choose File > Close from the menu at the top of the Fusion window.
- Click the Close X icon on the right of the composition's tab.

If the composition has unsaved changes, a dialog box appears allowing you to save before closing.

TIP: Compositions that have unsaved changes will display an asterisk (*) next to the composition's name in the Fusion Studio title bar and in the composition's tab.

To save the current composition, you can do the following:

- Choose File > Save and enter a name if the comp has yet to be named.
- Choose File > Save As to save under a new name.
- Choose File > Save Version to save the current composition with an added three-digit version number at the end of the name. Each time you save a version, the number automatically increments and the comp file is saved in the same location as the first version.

Auto Save

Auto save automatically saves the composition to a temporary file at preset intervals. Auto saves help to protect you from loss of work due to power loss, software issues, or accidental closure.

To enable auto save for new compositions, choose Fusion Studio > Preferences, and then locate Global > General > Auto Save in the Preferences dialog.

An auto-save file does not overwrite the current composition in the file system. A file with the same name is created in the same folder as the composition but with the extension .autosave instead of .comp. Unsaved compositions will place the autosave file in the default folder specified by the Comp: path in the Paths panel of the Global Preferences.

If an auto-save file is present when Fusion Studio loads a composition, a dialog will appear asking to load the auto-saved or original version of the composition.

The Composition File Format

Composition files are saved as readable plain text files. Using plain text files to describe a composition makes it easier to integrate Fusion into structured, visual effects pipeline and asset management solutions.

Composition files can be opened and edited using any standard text editing program. However, it is never a good idea to open the file using a word processor, such as Microsoft Word or Apple Pages, as these will generally save additional formatting information which will make the composition unreadable to Fusion.

Importing and Exporting a Composition from DaVinci Resolve

Although the compositions created in DaVinci Resolve's Fusion page are saved in the DaVinci Resolve project library as .drp project files, you can import and export Fusion composition files when in the Fusion page. This makes it very easy to share Fusion compositions between the different applications.

To export a Fusion composition from DaVinci Resolve:

- 1 From within DaVinci Resolve, switch to the Fusion page with the composition you want to export.
- 2 Choose File > Export Fusion Composition.
- 3 A Save dialog appears in which you can enter a name and location from the exported Fusion composition.

A .comp extension is added to the end of the filename. Only the node tree created in the Fusion page is exported. Clips not added to the Node Editor will not appear in the Fusion Studio bins. ResolveFX added to the comp will also not translate from the Fusion page to Fusion Studio.

MediaIn nodes from DaVinci Resolve are automatically converted to Loader nodes, and if the file path remains identical, the media is automatically relinked.

MediaOut nodes are converted to Saver nodes.

The return trip can also be performed, saving a composition file from Fusion Studio and importing it into the Fusion page within DaVinci Resolve.

To import a composition from Fusion Studio into the Fusion page within DaVinci Resolve:

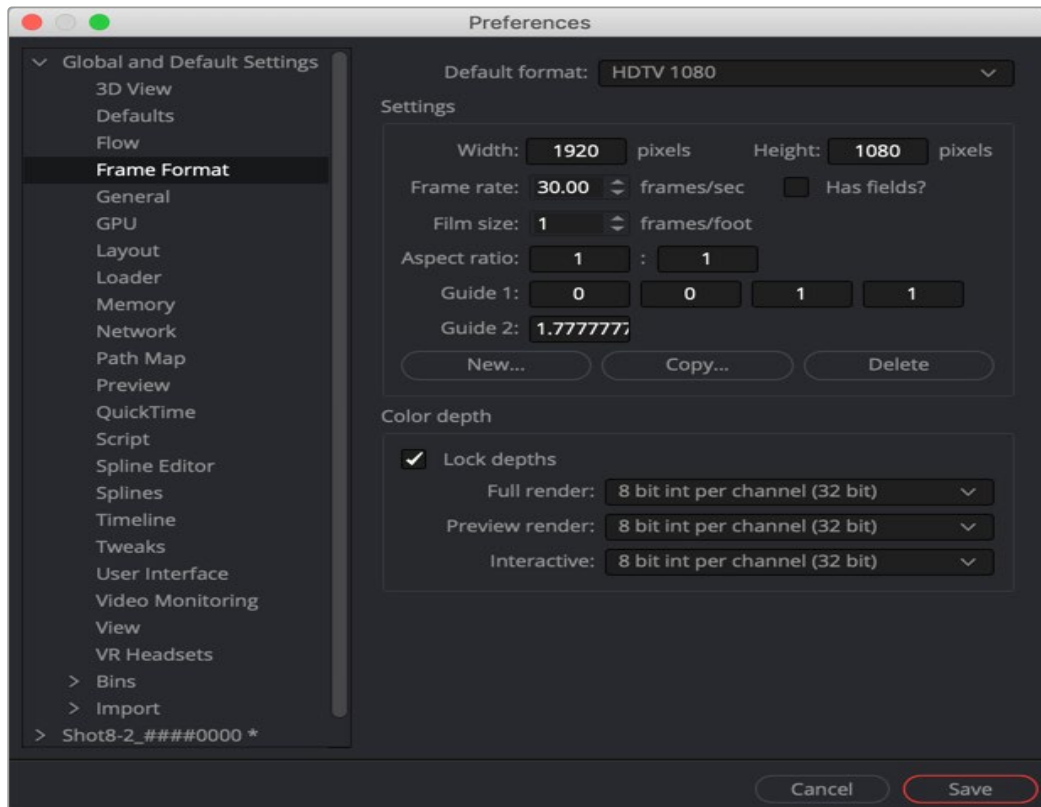
- 1 From within Fusion Studio, open the composition you want to move into the Fusion page.
- 2 From within DaVinci Resolve, switch to the Fusion page with an empty composition. The composition you import will completely replace the existing composition in the Fusion page Node Editor.
- 3 Choose File > Import Fusion Composition.
- 4 In the Open dialog, navigate to the Fusion comp and click Open.
- 5 The new comp is loaded into the Node Editor, replacing the previously existing composition.

TIP: To keep an existing comp in the Fusion page and merge a new comp from Fusion Studio, open Fusion Studio, select all the nodes in the Node Editor, and press Command-C to copy the selected nodes. Then, open DaVinci Resolve and switch the Fusion page with the composition you want, click in an empty location in the Node Editor, and press Command-V to paste the Fusion Studio nodes. Proceed to connect the pasted node tree into the existing one using a Merge or Merge 3D node.

Setting Up a Composition

Source media can come in a variety of formats, including HD, UHD, and 4K or larger. Often you will have different formats within a single comp. Each format has different properties, from resolution to color depth and gamma curve. Fusion can mix and match material of different formats together in a single composite, but it is important to note how Fusion Studio configures and combines materials of different formats when loading and merging them together.

When you open Fusion Studio, an empty composition is created. The first thing you do when starting on a new composition is to set the preferences to match the intended final output format. The preferences are organized into separate groups: one for global preferences, and one for the preferences of the currently opened composition.



Frame Format preferences for images generated in Fusion.

Although the final output resolution is determined in the Node Editor, the Frame Format preferences are used to determine the default resolution used for new Creator tools (i.e., text, background, fractals, etc.), aspect ratio, as well as the frame rate used for playback.

If the same frame format is used day after day, the global Frame Format preferences should match the most commonly used footage. For example, on a project where the majority of the source content will be 1080p high definition, it makes sense to set up the global preferences to match the frame format of the HD source content you typically use.

To set up the default Frame Format for new compositions, do the following:

- 1 Choose Fusion Studio > Preferences.
- 2 Click the Global and Default Settings disclosure triangle in the sidebar to open the Globals group.
- 3 Select the Frame Format category to display its options.

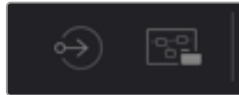
When you set options in the Global Frame Format category, they determine the default frame format for any new composition you create. They do not affect existing compositions or the composition currently open. If you want to make changes to existing compositions, you must open the comp. You can then select the Frame Format controls listed under the comp's name in the sidebar.

For more information on preferences, see *Chapter 75, "Preferences,"* in the DaVinci Resolve Reference Manual or *Chapter 15* in the Fusion Reference Manual.

Reading Clips into Fusion Studio

Once the Frame Format preferences are set, you usually begin to create a composite by reading in source media. When Fusion reads in media, it doesn't convert or move the original files; it simply reads the files in place, from whichever storage volume they're on. You are always dealing with the original source files in their original location.

Source media is read into a comp using a Loader tool. Although there are other tools within Fusion Studio that you can use to generate images like gradients, fractals, or text, each still image, image sequence, or movie file must be added to your comp using a Loader tool.



Loader and Saver tools are used to add media to Fusion Studio and render it out.

To add media to your comp, do one of the following:

- Click Effects to open the Effects Library, and then select Tools > I/O > Loader.
- Click the Loader icon in the toolbar.
- Right-click over the Node Editor, and then choose Add Tool > I/O > Loader.
- Drag a file from an OS file browser window into the Node Editor.

If multiple files are dragged into the Node Editor, a separate Loader is added for each file. However, if you drag a single frame from an image sequence, the entire series of the image sequence is read into the comp using one Loader, as long as the numbers are sequential.

To add only one frame of an image sequence to your comp:

- Hold Shift while you drag a single frame from an image sequence into the Node Editor.
This comes in handy when you want to read in photographs from a digital camera that are numbered sequentially.

A Loader represents any clip, image file, or graphic that you bring into Fusion. However, other types of media can also be brought into Fusion Studio. Photoshop PSD files, SVG splines, and 3D models in the Alembic, FBX, and OBJ format can be imported using the File > Import menu.

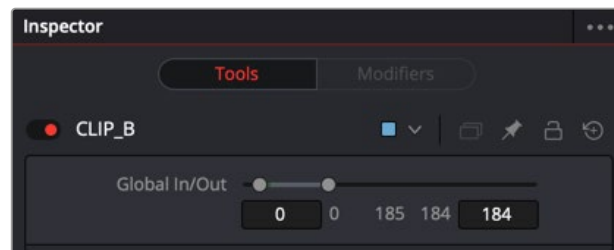
TIP: Using File > Import > Footage creates a new composition along with a Loader node for the footage. The selected media is automatically used for the name of the composition.

For more information about the Loader node, see *Chapter 105, "I/O Nodes,"* in the DaVinci Resolve Reference Manual or Chapter 45 in the Fusion Reference Manual.

Aligning Clips in a Fusion Studio Composition

When you add a clip into a composition, the resulting Loader node is added at frame 0 of the composition. However, the vital portion of the clip you are interested in may not start until a few frames or even seconds later. To ensure you can align the timing of each piece of media, each Loader includes timing and trimming options in the Inspector. You can also hold the first or last frame for a longer duration than the original media, and reverse or loop the clip to get more range for your composition.

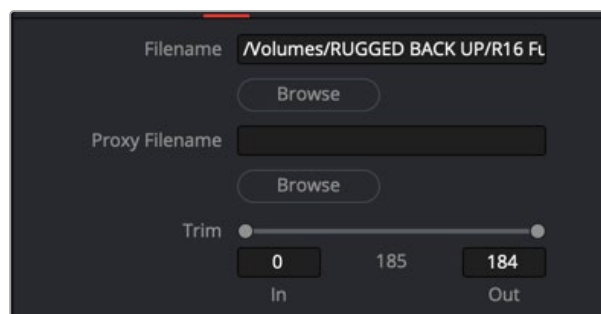
At the top of the Inspector are the Global In and Global Out settings. This range slider determines when in your composition the clip begins and ends. It is the equivalent of sliding a clip along a track in a Timeline. The Hold First Frame and Hold Last Frame dials at the bottom of the Inspector allow you to freeze frames in case the clip is shorter than the composition's global time.



Use the Global In/Out range slider to slide a clip in time to have it appear at the correct time in a comp.

Below the filename in the Inspector is a Trim In and Out range slider. This range slider determines the start frame and end frame of the clip. Dragging the Trim In will remove frames from the start of the clip, and dragging the Trim Out will remove frames from the end of the clip.

Although you may remove frames from the start of a clip, the Global In always determines where in time the clip begins in your comp. For instance, if the Loader has a Global In starting on frame 0, and you trim the clip to start on frame 10, then frame 10 of the source clip will appear at the comp's starting point on frame 0.



Use the Trim In/Out to remove unnecessary frames from the start or end of a clip.

Instead of using the Inspector to adjust timing, it is visually more obvious if you use the Keyframes Editor. For more information on the Keyframes Editor and adjusting a clip's time, see *Chapter 71, "Animating in Fusion's Keyframes Editor,"* in the DaVinci Resolve Reference Manual or *Chapter 9* in the Fusion Reference Manual.

Loader Node Inputs

Loader nodes have one Effects mask input and one output. In the case of the Effects mask input, connecting a mask node such as a Polygon or B-Spline node automatically creates an alpha channel in the Loader node.

TIP: If you connect a Mask node without any shapes drawn, that mask outputs full transparency, so the result is that the image output by the MediaIn node is blank. If you want to rotoscope over a MediaIn node, first create a disconnected Mask node, and with the Mask node selected and the Media In node loaded into the viewer, draw your mask. Once the shape you're drawing has been closed, connect the Mask node to the MediaIn node's input, and you're good to go.

Using Proxies for Better Performance

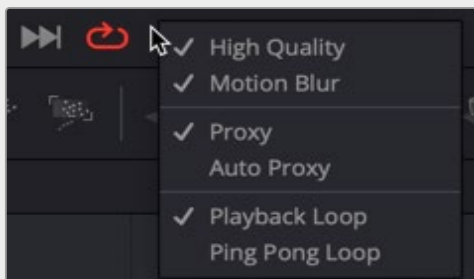
For increased performance, you can do one of two things:

- Generate smaller media files and write them to disk using Optimized Media in DaVinci Resolve
- Render out proxy files using Saver nodes in Fusion Studio

Both applications also allow you to generate proxies on-the-fly without rendering new files to disk using the Proxy and Auto Proxy options in the transport controls area.

To enable the Proxy and Auto Proxy options, you can do one of two things, depending on the version of Fusion you are using:

- In the Fusion page, right-click the empty area behind the transport controls to enable the Proxy option.



Proxy and Auto Proxy options in the transport controls right-click menu.

- In Fusion Studio, click the Proxy (Prx) button in the transport area to enable the usage of proxies.

The Proxy option reduces the resolution of the images as you view and work with them. Instead of displaying every pixel, the Proxy option processes one out of every x pixels interactively. In Fusion Studio, the value of x is determined by right-clicking the Prx button and selecting a proxy ratio from the drop-down menu. For instance, choosing 5 from the menu sets the ratio at 5:1. In the Fusion page, the proxy ratio is set by choosing Fusion > Fusion Settings and setting the Proxy slider in the General panel.



The Proxy menu sets the ratio for skipping pixels when processing.

The Auto Proxy button enables Fusion to interactively degrade the image only while adjustments are made. The image returns to normal resolution when the control is released. Similar to the Prx button in Fusion Studio, you can set the Auto Proxy ratio by right-clicking the APrx button and choosing a ratio from the menu.

When a Loader node is selected, the Inspector includes a Proxy Filename field where you can specify a clip that will be loaded when the Proxy mode is enabled. This allows smaller versions of the image to be loaded to speed up file I/O from disk and processing. This is particularly useful when working with high resolution files like EXR that might be stored on a remote server. Lower resolution versions of the elements can be stored locally, reducing network bandwidth, interactive render times, and memory usage.

The proxy clip that you create must have the same number of frames as the original clip, and if using image sequences, the sequence numbers for the clip must start and end on the same frame numbers. If the proxies are the same format as the original files, the proxies will use the same format options in the Inspector as the originals.

Presetting Proxy Quality

When using Fusion Studio, rather than right-clicking over the Proxy button to set the proxy quality, you can preset the standard and Auto Proxy quality in the Fusion Preferences window. The General pane in the Preferences window includes sliders for both standard Proxy files and Auto Proxy files. These sliders designate the default ratio used to create proxies when the Proxy and Auto Proxy modes are turned on. These settings do not affect the final render quality.

TIP: Even though the proxies are being processed smaller than their original size, the viewers scale the images so they refer to original resolutions.

File Format Options

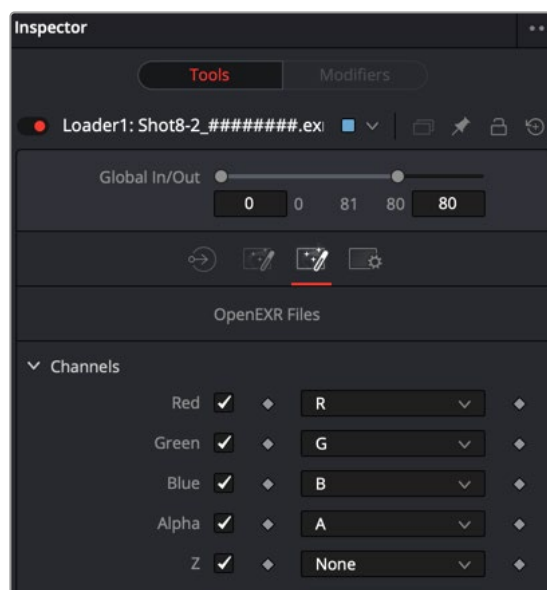
The Fusion interface in DaVinci Resolve and Fusion Studio display specific options for various file formats in slightly different ways. Where Fusion Studio displays most file-specific options in the Loader's Format tab, the Fusion page in most cases displays these options in the main Image tab of the MediaIn node. The only exception being the OpenEXR format. Its extensive options are displayed in a separate tab even in the MediaIn node of the Fusion page. Not all file formats have options. Only the DPX, OpenEXR, PSD, and QuickTime formats provide additional options when loaded.

DPX

The Format tab in Fusion Studio's Loader node for DPX files is used to convert image data from logarithmic to linear. These settings are often left in bypass mode, and the Log to Linear conversion is handled using a Cineon Log node.

OpenEXR

The OpenEXR format provides a compact and flexible high dynamic range (float) format. The format supports a variety of extra non-RGBA channels and metadata. These channels can be viewed and enabled in the Format tab of the Inspector.



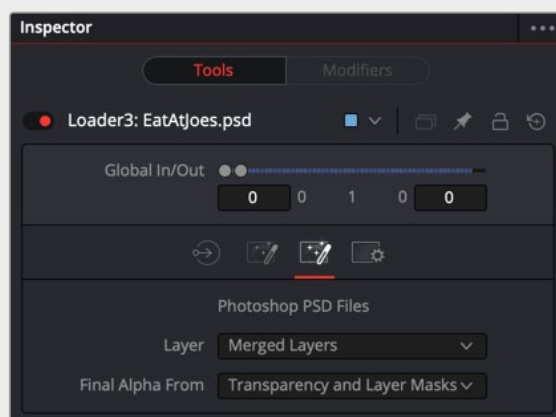
The Format tab in a Loader node Inspector displays Aux channels in EXR files.

Photoshop PSD

There are two methods for importing Photoshop PSD files. You can either import the PSD file and have it represented as a single node in the Node Editor or import the PSD and have each layer represented as a node in the Node Editor. If you do not need independent control over each layer or blend modes are not used when creating the PSD file, then importing the PSD file as a single node will make for a more manageable experience. If you do need control over each layer or Blend modes used in the PSD file are critical, then you should import the file so each layer becomes a node in the Node Editor. Each method is explained below for the Fusion page and Fusion Studio.

Using the Media Pool in DaVinci Resolve's Fusion page: Any PSD file added to the Media Pool in DaVinci Resolve can be accessed from the Fusion page. After dragging the PSD file from the Media Pool into the Node Editor, the image appears as a MediaIn node. From there, you can select which layer to use from the PSD file from the Layer drop-down menu in the Inspector.

Using a Loader node in Fusion Studio: This lets you read in Photoshop PSD files with the ability to select the layer in the PSD file that is used in the comp. Fusion can load any one of the individual layers stored in the PSD file, or the completed image with all layers. Transformation and adjustment layers are not supported.



The Format tab in the Inspector displays specific controls for a Photoshop PSD file.

To load all layers individually from a PSD file, with appropriate blend modes, do one of the following:

- In DaVinci Resolve, switch to the Fusion page and choose Fusion > Import > PSD.
- In Fusion Studio, choose File > Import > PSD.

Using either of the methods above creates a node tree where each PSD layer is represented by a node and one or more Merge nodes are used to combine the layers. The Merge nodes are set to the Apply mode used in the PSD file and automatically named based on the Apply mode setting.



The two layers of a Photoshop PSD file are imported and connected to a Merge node set to a Screen Apply mode.

QuickTime

QuickTime files can potentially contain multiple tracks. You can use the Format tab in the Inspector to select one of the tracks.

Loading Audio WAV Files in Fusion Studio

You can load WAV format audio-only files into Fusion Studio. The entire WAV file is loaded into RAM in order to quickly display the waveform in the Keyframe Editor. That being the case, it is best to use the shortest possible audio file you need for the Comp, so as not to use up more memory than necessary.

TIP: AIFF files can be loaded on macOS.

You can either load the audio file independent of any nodes, or load an audio file into the Saver node. The benefit of using a Saver node to load the audio is that you can view the audio waveforms in the Keyframes Editor.

To load a WAV audio file, do the following:

- 1 Right-click over the speaker icon and select Choose from the contextual menu.
- 2 In the file browser window, select the audio WAV file track to be used.

To load a WAV audio file using a Saver node, do the following:

- 1 Add a Saver node to the Node Editor.
- 2 In the Inspector, click the Audio tab and click the Browse button.
- 3 In the file browser window, select the audio WAV file track to be used.

To view the Audio Waveform in Fusion Studio, do the following:

- 1 Open the Keyframes Editor.
- 2 Expand the Saver track to view the audio waveform.

When you want to find the precise location of an audio beat, transient, or cue, you can slowly drag over the audio waveform to hear the audio.

Chapter 66

Rendering Using Saver Nodes

This chapter covers how to render compositions using Saver nodes in Fusion Studio and the Fusion page in DaVinci Resolve. It also covers how to render using multiple computers over a network when using Fusion Studio.

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